Eye as a Camera in XR: Reading Images, Colours and Eye Strain from a Consumer EEG Headband and a Consumer VR Headset

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***Abstract*—Mann showed in 2019 that the eye can be used as a camera: a square flickers at one position after another, the subject follows it, and the steady-state visual evoked potential (SSVEP) says how bright each position was. We move the method onto hardware a community health worker could carry: a Muse 2 headband with one cup electrode added over the visual cortex, and a Meta Quest 3S running the stimulus in its browser. We read a legible word in 18 minutes ($r = 0.91$ against time-shifted nulls of 0.23), eight colours as three calibrated colour planes (24 of 32 cells, reproduced over three sittings), and an acuity optotype drawn at random and sealed before the scan, named correctly from the EEG alone. The same recording gives an objective eye-strain reading: blinks stop while a cell flickers and burst when it goes dark, and the blink rate doubled between two sittings of one evening. We frame the result as a literacy-free visual screener for mobile health, with a prompt to rest the eyes as its first intervention. All results are from one subject.**

***Index Terms*—SSVEP; EEG; extended reality; consumer electronics; brain-computer interface; mobile health; visual screening; eye strain; just-in-time adaptive intervention; wearable computing.**

I. Introduction

A camera needs a lens, a sensor and a readout. The eye has the first two; Mann's "eye as a camera" [1], [2] supplied the third. A square flickers at a fixed rate, the subject looks at it, and the visual cortex answers with an SSVEP whose strength follows the light the square puts into the fovea [3], [4]. Scan the square across an image and the sequence of strengths is the image. The 2019 paper read "NO CAMERAS" and a portrait with a laboratory amplifier and 85 minutes of scanning.

This paper asks what the method can do on hardware that costs less than a phone and needs no technician: a Muse 2 headband (about $250) with one cup electrode through its auxiliary input, a Meta Quest 3S (about $300) running the stimulus as a web page, and a laptop. Our interest comes from the mobile health tradition of low-cost sensing in the hands of community health workers [5]–[7]. A visual screen read from the EEG asks the subject to name no letter and press no button; they look at a dot. The same recording carries what the mobile health literature has long wanted from wearables, an objective reading of the subject's state [8], [9], here the eye strain that flicker produces.

We contribute: the first SSVEP imaging on a consumer VR headset with a consumer headband; colour by three calibrated planes; a sealed, objective acuity read-out; eye strain from the same recording; and the field engineering that made the method survive outside a laboratory, with a synthetic subject as regression test.

II. Background

The SSVEP is largest over the occipital midline (Oz) and strongest for stimuli in the central field [3], which is what makes a single flickering patch under the gaze one pixel per position [1]. SSVEP interfaces in head-mounted displays are established [10]; imaging on one, and on the Quest family, is not.

Metaveillography of the eye. The eye-as-camera idea grew out of a longer programme on seeing sight itself. Mann photographed a camera's capacity to see with a light that brightens when the camera sees it, the sightfield, a time-reversed lightfield captured by long exposure [11], and named the sensing of sensors and of their capacity to sense metaveillance [12]; the photographs are metaveillographs, the measurements metaveillogrammetry [13]. Janzen and Mann turned the method on the human eye, measuring how far its sensing extends through space as a biological veillance flux [14], [15], and time-integrating that flux on objects as a veillogram [16]. The 2019 paper let the brain report instead of the subject: an LED carried through the visual field by a plotter, its brightness set by the SSVEP from a Muse headband, photographed over a long exposure into a metaveillograph of the eye's own receptive field [1], [2]. The same group then mapped the visual field into a "vidmap" [17] and read acuity from the SSVEP [18]. Two findings there carry this paper's health claims: the veillance field narrows when the observer ignores the flicker and widens when they track it, so it measures attention and not only optics, proposed in 2019 as a biomarker for attention problems in ADHD and ageing [1]; and the read-out is evidence rather than testimony, since nobody has to say what they see. Chromatic VEPs differ by colour axis and temporal frequency [19], which is why our colour planes are calibrated separately. The Muse headband is validated for event-related potentials [20] but has no electrode near Oz; the added cup is what makes imaging possible (cup alone $r = 0.92$, ear electrodes alone 0.36). A VR headset over the Muse improves the contact rather than degrading it [21].

Fletcher's group has built a decade of diagnostics around sensors a phone already has, for workers who are not special-

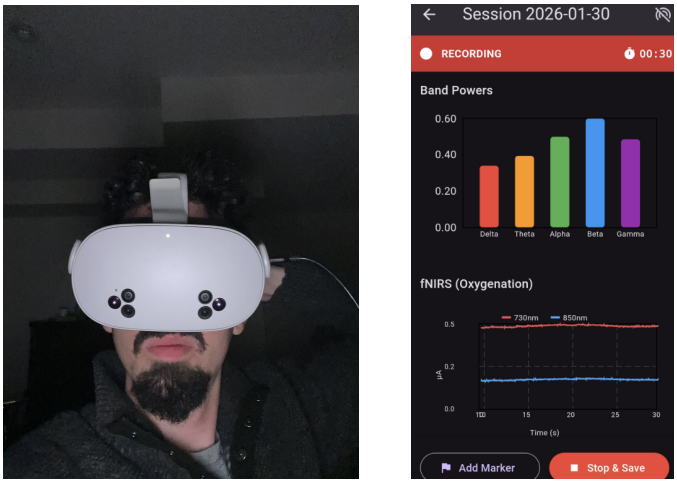


Fig. 1. Left: the Quest 3S worn over the Muse headband, the gold cup on the Muse auxiliary input held at Oz at the back of the head; the subject is the first author. Right: MuseLog on the phone during a recording: band powers and fNIRS live, markers, and the OSC stream of the raw EEG to the laptop.

ists: pulmonary screening from cough and lung sounds [7], [22], [23], child health tools [6], printed diagnostics read by the camera [24], thermal imaging with deep learning [25], [26], blood pressure from photoplethysmography [27], and hand-vein patient identification [28]. Vision screening is a gap in that list: the Snellen chart and the Ishihara plates need a literate, cooperative subject who reports what they see. The second thread is the wearable as a window on state, from iCalm [8] through sensor-plus-app systems for cognitive behavioural therapy [29]–[31] to the just-in-time adaptive intervention [32].

III. System

Hardware. Muse 2 at 256 Hz over Bluetooth to a phone, forwarded to the laptop over UDP; a gold cup with paste at Oz on the Muse auxiliary input, held by the headset strap; a Quest 3S; a laptop (Fig. 1). Under $600, nothing modified. The phone runs MuseLog [33], the lab's acquisition app for Android and iOS: it pairs with one or several Muse headbands, records the raw EEG with the auxiliary channel at 256 Hz, band powers, markers and, on the Muse S Athena, fNIRS, and streams it all live over OSC on UDP. A full-rate raw stream was added for this work: the app's decimated stream folded 34 to 49 Hz onto the SSVEP band.

Stimulus. A web page served by the laptop and opened in the headset's browser. One square flickers between black and white, or black and a colour, at the target position; a dot marks it and the subject follows the dot. The page flips on the display's frame clock, so at 72 fps the 12 Hz flicker is exactly three frames on and three off, and 9 Hz four and four; it reports the delivered frequency and its visibility, since a background tab renders at one frame per second.

Session. An eyes-closed alpha check confirms the cup is on the scalp. Calibration is six blocks of 8 s full-field flicker on and off; a line statistic at the flicker frequency with an exact permutation test, and a d' per electrode, decide whether to

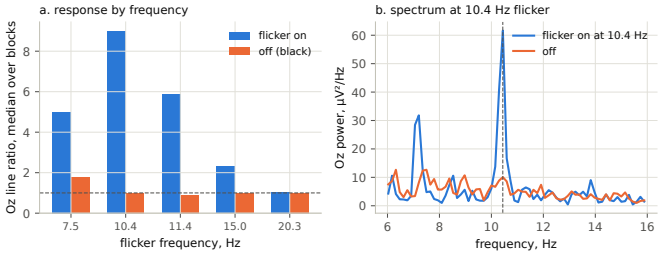


Fig. 2. The subject's response by frequency on the laptop (left) and the Oz spectrum at 10.4 Hz flicker (right): a single-bin line with the alpha hump unchanged.

proceed and how to weight the electrodes. The scan visits every grid position in serpentine order, 8 s each, with rest breaks on the subject's trigger, then revisits the positions whose black-or-white decision is uncertain. Each visit's pixel is the paper's whole-dwell ratio, power at the flicker frequency and its harmonic over the 14 to 50 Hz floor, weighted across electrodes; visits are averaged and the picture thresholded by Otsu's method in the log domain [34]. Every picture is scored against the target by correlation and against nulls in which the EEG is shifted in time by tens of seconds.

Colour by planes. Red, green and blue are three scans, each black-versus-colour with its own calibration; the three threshold decisions per cell give one of eight colours. Blue is weak (about 0.4 of red in power at the same contrast), so it gets a frequency sweep before the scan (6, 7.2, 9, 12 Hz; 9 Hz won on every sitting), a patch two cells wide, and a second visit to its ten least certain cells. Green gets a second pass; red, which has never missed a cell, one.

Guards. Five failures each cost a session before they were caught, from a laptop entering standby on battery to an ear electrode loose from minute 11; the driver now guards against four, and a loose-channel guard is next. A synthetic subject that emits the right SSVEP for the stimulus log exercises the whole stack without a person.

IV. Results

All results are from one subject, the first author, September to October 2026.

Where the eye answers. The first sweep (Fig. 2) put the response between 7.5 and 12 Hz: Oz line ratios of 9.0 at 10.4 Hz and 5.9 at 11.4 Hz against 1.0 with the screen off, nothing at 20 Hz. Mann's 15 Hz sits on the weak side of this curve, so our sessions use 12 Hz, the nearest frequency the display delivers exactly. A stock Muse without the cup showed no response. Measure the subject first.

A legible word. "NO" in a pixel font on 9 by 5 positions (Fig. 3): $r = 0.79$ after one pass, 0.88 after two, 0.91 after three, nulls at most 0.23, 43 of 45 positions right after the threshold, 18 minutes. A square 4.6 times larger reached 0.87 in two passes, so a larger square did not buy signal.

Eight colours over three sittings. Table I and Fig. 4. The first sitting read red 32, green 28 and blue 28 of 32, giving 24 of 32 cells in the right colour and the hue right on 19 of 24 coloured cells, $r = 0.79$ against nulls of 0.32. Two hours later,

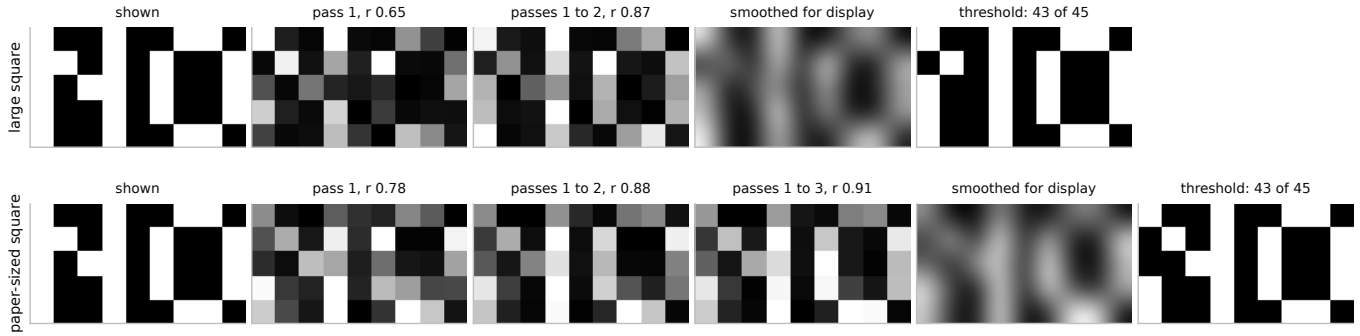


Fig. 3. "NO" read from the EEG in the Quest 3S, 12 Hz, 8 s per position, with a large square (top) and the paper-sized square (bottom): shown, after each pass, smoothed as in the 2019 paper, and thresholded. Both reach 43 of 45.

TABLE I
THE EIGHT-COLOUR CHART (8 BY 4) OVER THREE SITTINGS: CELLS RIGHT PER PLANE, CELLS IN THE RIGHT COLOUR, COLOURED CELLS WITH THE RIGHT HUE, AND COLOUR r WITH THE LARGEST TIME-SHIFTED NULL.

sitting	R	G	B	right	hue	r (null)
2026-10-10 17:57	32	28	28	24/32	19/24	0.79 (0.32)
2026-10-10 19:50	29	31	23	20/32	16/24	0.66 (0.32)
2026-10-12 10:40	32	31	23	23/32	16/24	0.75 (0.25)

20 of 32: green's second pass helped, blue's second pass was lost to the right ear electrode coming loose at minute 11. Two days later, 23 of 32, red perfect again, blue lost on the right half of the field with every electrode holding. One wrong cell in four on the chart is almost entirely blue's.

A sealed acuity read-out. A tumbling E and a ring that is closed (O) or missing its bottom (U) sit on a 12 by 7 grid; one of the eight combinations is drawn at random and sealed before the scan. A template correlation on the EEG picture names the orientation and the ring, then the seal is opened. On the first human run (Fig. 5) the picture reached $r = 0.89$, 80 of 84 cells, and the read-out said E up, ring U, with margins of 0.24 and 0.22 over the runners-up; the seal agreed. One trial at one-in-eight is a demonstration; a companion exam has since run the size series (about 20/18 and 20/25) [35], and this picture, printed as a lithophane, shows the optotype against a light.

Toward a real photograph. The 2019 portrait came back a silhouette. We prepare a sealed forced choice among four portraits cut to 16 by 16 cells at four grey levels, the block-portrait size at which a face is recognisable [36], [37]; on the clean targets the read-out names the right person every time with a quarter of the cells flipped, chance one in four. A flat illustration in the eight corner colours survives the three planes; a colour photograph, its colours mid-cube, does not.

A three-dimensional object. The planes scan reads any colour image at grid size, so an object's texture atlas can go through it and come back painted on the object. A die's six faces were unfolded into a 54-cell atlas of black, white and one red pip, and scanned at 5, 5 and 8 s per cell in one pass with the ten least certain blue cells shown again, 18 minutes in a rested morning sitting. Snapped to the die's three colours, 51 of 54 cells came back right (black 19 of 20, white 31 of 33, the pip in place) and the cube is painted from the EEG (Fig. 6); on the stricter rule used for the chart, right in all three planes at once, 35 of 54, with per-plane r of 0.66, 0.67 and 0.70 against a time-shifted null of 0.12. Printed in relief on a Bambu Lab H2C, white flush, black sunk and the pip raised by 1.2 mm, the die becomes an object (Fig. 6, bottom).

Eye strain from the same recording. After two colour sittings in one evening the subject reported irritated eyes, a slight forehead ache and no drowsiness. On a Muse under a headset the blink is best seen through the Fpz reference on the far electrodes, as a shared deflection on TP9, TP10 and the cup. Counted so, the blink rate was 6.8 per minute in the first sitting and 12.1 in the second against 3 to 4 on a reference day, with amplitude and duration unchanged. Within a sitting (Fig. 7) blinks nearly stop while a cell flickers and burst within a second of it going dark, 50 to 65 per minute in that second, a suppress-and-release pattern the reference day lacks [38], [39]. Drowsiness markers were flat: no rise in frontal theta, no long blinks; a rise in alpha seen earlier was the loose ear electrode. The forehead ache is not in the EEG (frontalis 1.5 to 2.3 μV rms); the headset's facial interface is the likelier cause. The SSVEP itself fell by half between the blue planes of the two sittings, and the cup's noise rose from minute 20 of the second, so part of what reads as fatigue is contact.

The ear as a microphone. The same headband reads sound. The first two paragraphs of *Alice's Adventures in Wonderland* (31.5 s of a public-domain LibriVox reading) were played 24 times from the headset's own speakers while the subject looked at a dot. The brain follows the loudness of speech, and a linear decoder can read that envelope back from scalp EEG [40], [41]. Ours is a ridge decoder on the five channels at lags of -125 to $+500$ ms reading the 1 to 8 Hz envelope; each play is decoded by a model trained on the others, with its first and last 3 s left out so onset and offset responses do not count, and the mean of the held-out reconstructions is the brain's envelope of the clip (Fig. 8). On a morning sitting with every electrode in contact it correlated with the true envelope at $r = 0.36$ over 23 plays (circular-shift null 95% 0.10, $p < 0.001$), one play at 0.086 (null 0.022), and with the EEG shifted off the clips at chance (0.035, $p = 0.29$); the night before, with the cup not in contact, 0.18. The signal is on the ear electrodes and the cup (TP10 0.20, AF8 0.16, Oz 0.15, TP9 0.14 alone), AF7 nothing. Research caps with 64 channels reach 0.10 to

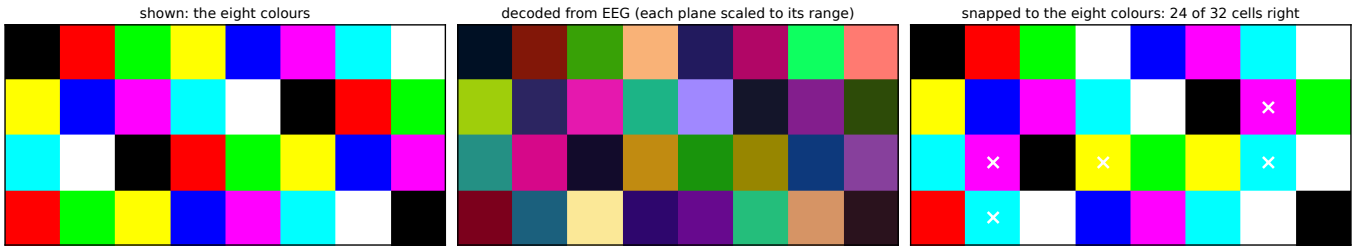


Fig. 4. Eight colours on the first colour sitting: shown, decoded with each plane scaled to its range, and snapped to the eight colours with wrong cells marked. 24 of 32.

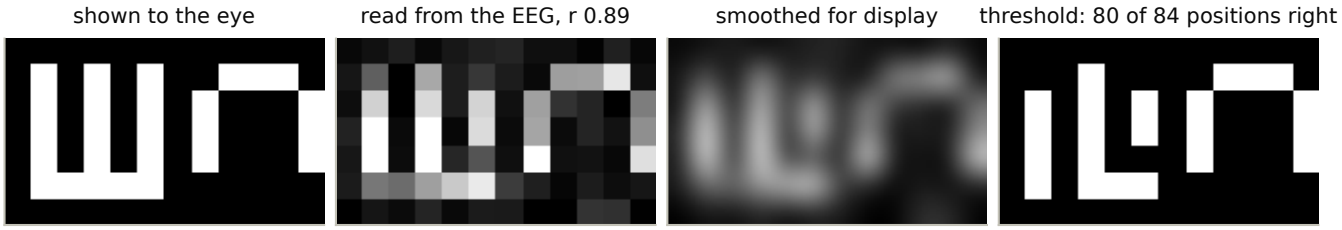


Fig. 5. The sealed pair read from the EEG: shown, one value per position, smoothed, thresholded. The read-out named E up and ring U before the seal was opened; both right.

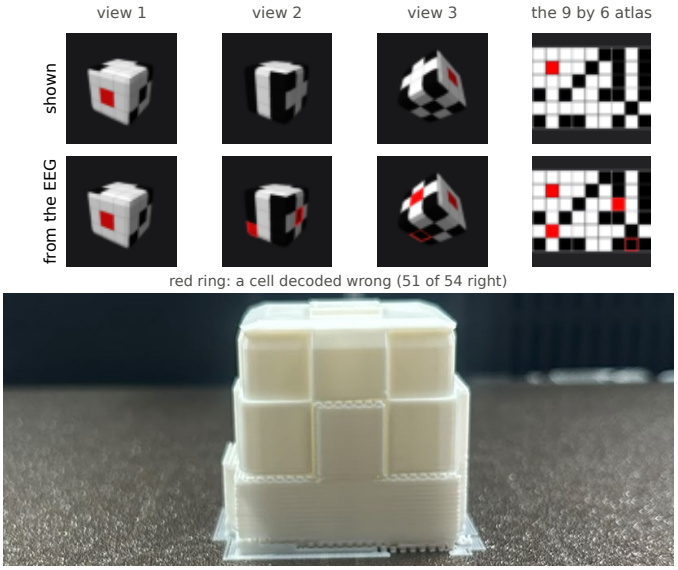


Fig. 6. A die read from the eye. Top: the cube as shown, three views, and the 9 by 6 atlas the subject scanned. Middle: painted with the colours decoded from the EEG, 51 of 54 cells right. Bottom: printed in relief on a Bambu Lab H2C.

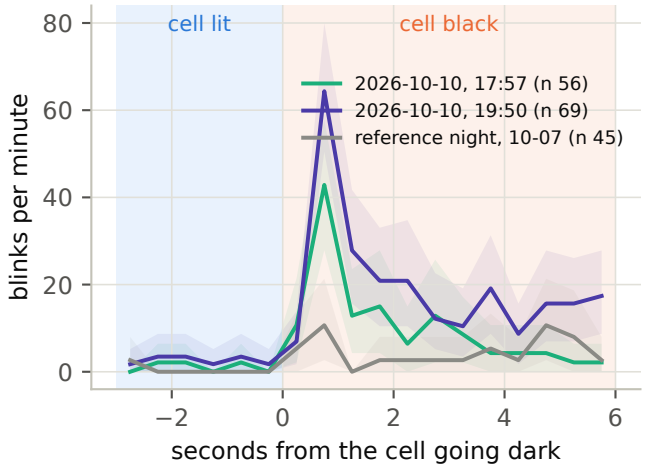


Fig. 7. Blink release: blinks per minute in half-second bins around the moment a flickering cell turns black, two colour sittings against a reference day; bands are 95% block-bootstrap intervals.

0.19 per play [41]; the headband recovers by repetition what it lacks in channels. The reconstruction is played back as sound. The speakers sit beside TP9 and TP10, so part of the response could be sound induced into the electrodes; a control with the headband on the desk and nobody wearing it, and a sitting with speakers across the desk, are pending.

Music through a diffusion model. Postolache et al. [42] reconstructed music from 128-channel laboratory EEG at a semantic level, timbre and mood rather than fine detail, by conditioning a frozen text-to-music model, AudioLDM2 [43], on the EEG through a ControlNet adapter [44]. We ran the same pipeline on the four Muse channels (TP9, AF7, AF8, TP10). One participant heard eight short stimuli in two blocks. Trials were cut to 3.84 s from onset (stimuli resampled to 16 kHz mono and zero-padded; silent controls excluded), median-centred, notch-filtered at 60 Hz, high-passed at 0.5 Hz, downsampled to 125 Hz and denoised with the Adaptive Chirplet Transform (ACT) [45]–[47], which models each channel in 2 s windows with a 1 s hop as a sum of Gaussian chirplets. A strided 1-D convolution maps the 4×480 EEG onto the $8 \times 96 \times 16$ audio latent, one EEG step per

TABLE II
MUSIC THROUGH A DIFFUSION MODEL, CROSS-BLOCK ($n = 40$ TRIALS PER BLOCK). CLAP: MEAN SIMILARITY TO THE TRUE STIMULUS; ID: EIGHT-WAY IDENTIFICATION RATE (CHANCE 0.125).

	Test block 02				Test block 03			
Model	CLAP	p	ID	p	CLAP	p	ID	p
No-EEG baseline	0.249	0.026	0.100	0.754	0.245	0.167	0.075	0.892
Classify-then-play	0.681	0.637	**0.250**	**0.023**	0.770	0.133	0.125	0.571
ControlNet (EEG)	0.496	0.327	0.100	0.754	0.468	0.532	0.100	0.754
Label-shuffle null	0.432	0.502	0.175	0.228	0.418	0.891	0.075	0.892

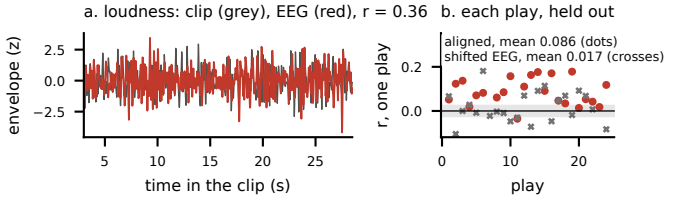


Fig. 8. The ear as a microphone. a: the loudness envelope of the 25.5 s scored from each play (grey) and the mean of the 23 held-out reconstructions from the EEG (red), $r = 0.36$. b: each play decoded by a model that never saw it, against the shift null (band) and the shifted-EEG control (crosses).

latent frame. Each model was trained from scratch on one block and tested on the other, then the roles swapped (1000 AdamW steps, learning rate 10^{-4}, batch 4; DDIM sampling, 25 steps), and scored as in [42] by CLAP similarity to the true stimulus [48] (4000-iteration permutation test) and by eight-way identification in CLAP space, against unconditioned AudioLDM2 (*no-EEG*), the same model trained on shuffled EEG–stimulus pairs (*label-shuffle*), and a band-power classifier that plays back its predicted stimulus (*classify-then-play*; Table II). The EEG-conditioned model came closer to the true stimulus than the label-shuffle control on both blocks (CLAP 0.496 vs. 0.432 and 0.468 vs. 0.418), consistent in direction but not yet significant with about 40 training trials per block; the classifier named the stimulus at twice chance on block 02 (25%, $p = 0.023$), so the four channels carry stimulus information. The pipeline runs end to end on a consumer headband; more sessions, participants and a frozen held-out evaluation are next.

V. TOWARD A SCREENING TOOL FOR THE FIELD

What a community health worker would do with this decides the design. The metaveillography lineage names the readings: a visual field, an acuity and an attention state, the last from how the field narrows and widens [1], [17], [18]. Three readings come from one sitting under half an hour. A *colour response profile*: the per-colour calibrations are three numbers before any scan, the raw material of an objective colour-vision screen once the stimuli are isoluminant. An *acuity read-out*: the sealed optotype, then a threshold series, giving acuity without a verbal response, as the sweep VEP does in a clinic [35]. An *eye-strain reading and a prompt*: the blink pattern and rate are live from the same headband, and a "rest your eyes" message is the simplest just-in-time intervention [32], [49].

Limits. One subject. One cell in four wrong on a 32-cell colour chart; a 96-cell picture is an hour. Blue is fragile, and 9 Hz sits next to alpha. Contact is the largest practical risk: the cup's d' ranged from 0.8 to 5.6 across days, and one loose electrode cost a plane. Flicker at 3 to 30 Hz can provoke seizures in photosensitive epilepsy and the method must not be used with anyone with that history. The eye-strain and ear results are one evening and two sittings, without controls. Next: two more subjects, a letter-chart calibration of the grating acuity, the loose-channel guard, the sealed photograph, the ear's desk control, the prompt.

VI. CONCLUSION

The eye as a camera works on a $250 headband with one added cup and a $300 headset: a legible word, eight colours over three sittings, and an optotype named from the EEG before its seal was opened, from a subject who only looked at a dot. The same headband is an ear as a microphone: the loudness of a spoken passage read back from the EEG at $r = 0.36$ and played as sound. The recording that gives the image also gives the eye strain. That is the start of a visual screener in the low-cost, objective, worker-operated tradition; the rest is subjects, stimuli that separate colour from brightness, and a prompt that closes the loop.

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